

Living With QUANTUM COMPUTING



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Importance of computers in the present day is well known to all. These machines have almost taken over manpower and mostly for the betterment (with the exception of creating unemployment). Still, people expect computers to be more useful and powerful in times to come, and different computing technologies of the future are always on constant watch.

The future of computing and the new fields of computer sciences paving the way for the next digital revolution are common topics of discussion. In this direction, quantum computing technologies and their emergence in the near future are discussed. It is expected that quantum computing technologies will reach the masses by 2020. This article presents how quantum computing will change lives, society, the economy and the entire working system.

Computing technologies, in general, are based on a series of assumptions, which are:

- A technological society could eventually achieve the capability of creating a computer simulation that is indistinguishable from reality to the inhabitants of the simulation.
- Such a society would not do this once or twice. These would create many such simulations.

- Left to run long enough, the societies within the simulations would eventually be able to create their own simulations, also indistinguishable from reality to the sub-simulations inhabitants.

Certain tasks, which have long been thought impossible (or intractable) for classical computers, will be achieved quickly and efficiently by quantum computers. These computers will be millions of times more powerful than conventional computers, and quantum computing could lead to huge improvements in machine learning, artificial intelligence, computer simulations and cryptography. All of this could fundamentally alter the way our society operates.

Quantum computers will be able to outperform conventional computers in the fields of machine learning (training computers to use data to, effectively, make decisions without additional human input, to run search engines, spam email filters, voice- or facial-recognition technologies or self-driving cars, for example) and simulation technologies.

What quantum computing is

Quantum computing is essentially harnessing and exploiting the amazing laws of quantum mechanics to process information. A traditional computer uses long strings of bits, which encode either 0 or 1. A quantum computer, on the other hand, uses quantum bits, or qubits.

A qubit is a quantum system that encodes 0 and 1 into two distinguishable quantum states. Qubits represent atoms, ions, photons or electrons and their respective control devices that work together to act as computer memory and a processor. But, because qubits behave quantum mechanically, we can capitalise on the phenomena of superposition and entanglement.

Superposition is the ability of a quantum system to be in multiple states at the same time, that is, something can be here and there, or up and down at the same time.

Known as D-Wave, Google's quantum computer is making the leap from 512 qubits to more than a 1000 qubits—a leap that does not require a significant increase in power

